Quantile-Initialised Ordinal Collaborative Filtering with Gated Auxiliary Fusion

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Abstract

We propose a simple calibration recipe for cumulative-link collaborative filtering: initialise the monotone thresholds from the empirical rating quantiles of the training set, parameterise their spacing through a softplus-inverse, and exclude them from L_2 regularisation. The recipe matches the data marginal exactly at the start of training and, on a transparent dot-product matrix-factorisation backbone, holds calibrated test negative log-likelihood (1.2584 across the five canonical MovieLens-100K splits, vs the $\log 5{\approx}1.61$ uniform baseline) without sacrificing RMSE. The recipe is paired with a per-dimension symmetric gated fusion of side information — a modest design choice that turns out to do most of the empirical work — and a manifold-projection diagnostic that exposes item-side genre structure that linear projections miss (2-D IsoMap silhouette $+0.23$ vs $+0.05$ under PCA). The combined gated+ordinal configuration attains RMSE 0.9124 ± 0.0036 across all five MovieLens-100K splits (beating the no-fusion baseline 0.9160 and the additive-fusion variant 0.9189 on the across-split mean and on four of the five individual splits), with the same ordering preserved on a $10\times$-larger MovieLens-1M scale-up (RMSE 0.8481 ± 0.0006, NLL 1.184). The contribution is implementation-level: the components are well established; the recipe makes them work together calibrated, with a small set of supporting diagnostics.

1 Introduction

Collaborative filtering (CF) for explicit feedback is canonically framed as predicting an unobserved scalar rating r_{ui} from a sparse user–item rating matrix $R \in \mathbb{R}^{n\times m}$ [19]. Latent-factor models based on matrix factorisation [9] dominate the public benchmarks on MovieLens [5] and have evolved into deep hybrids [7, 23] and factorisation-machine families [18, 6, 24, 4, 2, 13]. Two persistent issues remain when these models are applied as a course-style data-analytics tool rather than a production recommender. First, side information (user demographics, item genres) is typically injected by adding a projected feature vector to the latent embedding, with no learned mechanism to decide *when* to trust the embedding versus the side information. Second, the rating is treated as a continuous quantity even though it is in fact a five-class ordinal variable; the resulting predictions can be RMSE-competitive yet badly calibrated as distributions over star ratings.

Related work. The shortcomings above have each been addressed in isolation. *Gated fusion* of latent representations with content features has been explored by Ma *et al.*'s GATE [15], which uses a sigmoid gate to blend an item embedding with item content; gating is also implicit in attention-based factorisation [24]. *Cumulative-link ordinal regression* for recommendation has a long history starting with McCullagh's classical formulation [16], and was applied to MovieLens by Koren and Sill in OrdRec [10, 11]. Two recent papers operate in essentially the same setting as ours: OrdNMF [3] combines an ordinal head with non-negative matrix factorisation, and OPRFM [25] couples ordinal probit regression with factorisation machines on MovieLens-100K, 1M, and 10M. The cumulative-link parameterisation itself has seen renewed interest in deep ordinal classification [22, 1] and in ordinal random fields for recommenders [14]. Manifold visualisation of learned recsys embeddings is rare; standard tools include IsoMap [20], t-SNE [21], and UMAP [17]. Our contribution is therefore not the invention of any single component, but their integration on a transparent linear core, together with a calibration recipe (empirical-quantile threshold initialisation) and a geometry diagnostic.

Contributions. We frame the contribution as implementation-level rather than algorithmic; each component below is itself well known.

1. *Headline.* A simple calibration recipe for cumulative-link CF: initialise the monotone thresholds from the training-set rating quantiles via $\theta_k = \mathrm{logit}(\hat{F}_k)$, recover the unconstrained δ_j by inverting softplus, and exclude $\{\theta_1, \delta_j\}$ from L_2 regularisation. The predicted marginal then matches the data marginal at $s = 0$ to numerical precision and the head holds calibrated NLL throughout training.
2. *Supporting design choice.* A symmetric per-dimension gated fusion that blends ID embeddings with side-feature projections on both the user and the item sides, with zero-initialised gates so that training begins at the pre-fusion MF solution. Empirically this fusion does most of the RMSE work — the ordinal head's contribution is concentrated on calibration and per-class accuracy.
3. *Supporting diagnostic.* A manifold-projection lens (PCA, MDS, IsoMap) on the post-fusion embeddings, with silhouette scores stratified by occupation and dominant genre. IsoMap recovers item-side genre structure that linear projections miss (2-D silhouette $+0.23$ vs $+0.05$ under PCA), and the gap is itself an interpretable property of the linear core.

2 Problem Definition

Let $\mathcal{U} = \{1, \ldots, n\}$ be the set of users and $\mathcal{I} = \{1, \ldots, m\}$ the set of items. Observed ratings form a sparse set $\mathcal{D} = \{(u_t, i_t, r_t)\}_{t=1}^{N}$ with $r_t \in \{1, 2, 3, 4, 5\}$. Each user u carries a side-information vector $f_u \in \mathbb{R}^{d_u}$ (z-scored age, one-hot gender, one-hot occupation in MovieLens-100K, $d_u = 24$) and each item i carries $g_i \in \mathbb{R}^{d_i}$ (multi-hot genre indicator, $d_i = 19$). The task is to predict $\hat{r}_{ui}$ for held-out (u, i) pairs, evaluated by RMSE, MAE, classification accuracy of the rounded prediction, and where applicable the held-out negative log-likelihood (NLL) of the predicted rating distribution.

3 Method

3.1 Matrix factorisation backbone

We start from a biased dot-product MF model. Each user and item carries a d-dimensional embedding ($p_u, q_i \in \mathbb{R}^d$) and a scalar bias ($b_u, b_i \in \mathbb{R}$). The pre-head score is

$$s_{ui} \;=\; p'_u \cdot q'_i \;+\; b_u \;+\; b_i, \tag{1}$$

where p'_u, q'_i are the (possibly fused) representations defined below. With fusion disabled, $p'_u \equiv p_u$ and $q'_i \equiv q_i$.

3.2 Gated auxiliary fusion

Let $a_u = \mathrm{ReLU}(W_u f_u + b_{f,u})$ be a learned projection of the user side features into the embedding space, and similarly $a_i = \mathrm{ReLU}(W_i g_i + b_{f,i})$ for items. We define the fused user representation as

$$g_u \;=\; \sigma(W_g\,[\,p_u \,\|\, a_u\,] + b_g)\,, \qquad p'_u \;=\; (\mathbf{1} - g_u) \odot p_u \;+\; g_u \odot a_u, \tag{2}$$

where $\|$ denotes concatenation and $\odot$ is the Hadamard product, so the gate $g_u \in [0,1]^d$ acts *per dimension*. The item-side fusion is identical with (q_i, a_i) in place of (p_u, a_u). We zero-initialise both W_g and b_g, which makes $g_u \equiv \frac{1}{2}\mathbf{1}$ at training start. This anchors the optimiser at the equal-weight average of the pre-trained MF solution and the ReLU-projected side features — the gate then opens and closes per dimension as gradient flows. As an ablation we also consider the *additive* variant $p'_u = p_u + a_u$ and the *none* variant $p'_u \equiv p_u$ (and similarly for items).

3.3 Ordinal regression head with the calibration recipe

The headline contribution of this paper sits in this subsection: a small recipe for cumulative-link CF that holds the predicted rating distribution calibrated to the training-set marginal from step zero. The recipe has three ingredients — monotone-threshold parameterisation (Eq. 3–4), *empirical-quantile initialisation* (Eq. 5), and exclusion of the threshold parameters from L_2 regularisation. None of the three is individually new; their combination is what we contribute.

For the head we treat r as an ordinal variable with $K = 5$ classes. We use the cumulative-link model of McCullagh [16] with monotone thresholds $\theta_1 < \theta_2 < \theta_3 < \theta_4$:

$$P(r \le k \mid s) = \sigma(\theta_k - s), \qquad P(r = k \mid s) = \sigma(\theta_k - s) - \sigma(\theta_{k-1} - s), \tag{3}$$

with $\theta_0 = -\infty$ and $\theta_5 = +\infty$. We enforce monotonicity by the softplus reparameterisation

$$\theta_k = \theta_1 + \sum_{j=1}^{k-1} \mathrm{softplus}(\delta_j), \tag{4}$$

which guarantees $\theta_k < \theta_{k+1}$ for any unconstrained $(\theta_1, \delta_1, \ldots, \delta_{K-2})$. This trick has been used in deep ordinal classification [22, 1]. We use the logit link in (3); replacing the logistic CDF with the normal CDF (probit, as in OPRFM [25]) shifts headline RMSE by -0.0009 and NLL by $+0.005$ across the five splits, both within between-split standard deviation, so we treat the choice as a parameterisation detail rather than a methodological commitment.

Empirical-quantile initialisation (the recipe). Default zero initialisation (or a single $\theta_1 = -2$) is poor: at $s = 0$ the predicted marginal puts almost all mass on the lowest class, and the model spends several epochs warming up. We instead initialise from the empirical rating distribution. Let $\hat{p}_k$ be the training-set frequency of rating k and $\hat{F}_k = \sum_{j \le k} \hat{p}_j$ its cumulative. Choosing

$$\theta_k = \mathrm{logit}(\hat{F}_k), \qquad \delta_j = \log\big(\exp(\theta_{j+1} - \theta_j) - 1\big) \tag{5}$$

guarantees $P(r{\le}k \mid s{=}0) = \hat{F}_k$, so the predicted marginal at the start of training equals the data marginal exactly. The second equation is just the softplus inverse, recovering the unconstrained δ_j that the optimiser then updates. Together with excluding $\{\theta_1, \delta_j\}$ from weight decay — their scale carries calibration information that L_2 shrinkage destroys — this is the calibration recipe we propose. It is ~ 10 lines of code, agnostic to the score s, and we use it on every ordinal cell in the experiments below.

The training loss is the negative log-likelihood $\mathcal{L}_{\mathrm{ord}} = -\frac{1}{N} \sum_t \log P(r = r_t \mid s_{u_t i_t})$, and the point prediction is the expected rating $\hat{r}_{ui} = \sum_{k=1}^{5} k\, P(r = k \mid s_{ui})$. As an ablation we also consider the standard *sigmoid* head: $\hat{r}_{ui} = 4\sigma(s_{ui}) + 1$ trained with mean-squared error.

Algorithm 1 Forward pass + ordinal NLL for one observed rating (u, i, r).

$a_u \leftarrow \mathrm{ReLU}(W_u f_u + b_{f,u});\ a_i \leftarrow \mathrm{ReLU}(W_i g_i + b_{f,i})$ ▷ side projections

$g_u \leftarrow \sigma(W_g^{(u)}[E_u[u] \| a_u] + b_g^{(u)});\ g_i \leftarrow \sigma(W_g^{(i)}[E_i[i] \| a_i] + b_g^{(i)})$ ▷ per-dim gates

$p' \leftarrow (\mathbf{1} - g_u) \odot E_u[u] + g_u \odot a_u;\ q' \leftarrow (\mathbf{1} - g_i) \odot E_i[i] + g_i \odot a_i$

$s \leftarrow p' \cdot q' + b_u + b_i;\ \theta_k \leftarrow \theta_1 + \sum_{j<k} \mathrm{softplus}(\delta_j)$

$P(r{=}k \mid s) \leftarrow \sigma(\theta_k - s) - \sigma(\theta_{k-1} - s)$, with $\theta_0 = -\infty, \theta_K = \infty$

return $\mathcal{L} = -\log P(r{=}r_{\text{true}} \mid s)$

3.4 Complexity

The MF backbone has $(n + m)d$ embedding parameters plus $n + m$ biases; for MovieLens-100K with $n = 943$, $m = 1682$, $d = 128$, that is $\approx 3.4 \times 10^5$ parameters. Each fusion module adds a projection $(d_{u/i} \cdot d + d)$ and a gate $(2d^2 + d)$; summed over both sides this is $\approx 7 \times 10^4$ parameters — about one-fifth of the embedding budget. The ordinal head adds $K - 1 = 4$ parameters. Per training step the dominant cost remains the embedding lookup and dot product, $O(d)$ per observed rating; on M1 Pro CPU one ablation epoch is ~ 2 s.

4 Experiments

4.1 Setup

We use MovieLens-100K [5] on its five canonical $80/20$ splits `u1`–`u5`, each yielding $80{,}000$ training and $20{,}000$ test ratings on the same 943 users and $1{,}682$ items. Headline numbers are reported as mean $\pm$ std across the five per-split means (each split-mean itself averages three training seeds), giving 15 trained models per ablation cell. The visualisation (§4.5) and cold-start (§4.6) experiments are restricted to the `u1` split. User side features are z-scored age, two-class one-hot gender, and 21-class one-hot occupation ($d_u = 24$); item features are 19-genre multi-hot ($d_i = 19$). All experiments use Adam [8] with learning rate 10^{-3} and weight decay $\lambda = 10^{-5}$ on all parameters except the ordinal-head thresholds. We carve a 10% random slice of `u1.base` (8,000 ratings) as a held-out validation set, leaving 72,000 for training; the val-split seed is fixed and shared across all training-seed replicates so val-split variance is not conflated with model-init variance. *Early stopping is driven by val-set RMSE only*, and the test set is read exactly once at the end of training, with the best-val weights restored. We report two protocols differing only in early-stopping aggressiveness: *headline* uses patience $= 30$ to allow the full 30-epoch budget, matching a fixed-budget training protocol of 30 epochs; *ablation* uses patience $= 10$ to keep the 18-run sweep tractable (~ 15 minutes wall on M1 Pro CPU). The two protocols agree on headline gated+ordinal RMSE to within ± 0.003.

4.2 Baselines and metrics

We compare against three transparent baselines: (i) truncated SVD with rank 20 on the mean-centred rating matrix, (ii) biased MF trained with MSE, and (iii) NMF, identical to MF but with the user/item factor matrices constrained to be element-wise non-negative via projected gradient descent (clipping to ≥ 0 after each Adam step, biases unconstrained, following Lee & Seung [12]). Metrics are RMSE, MAE, classification accuracy of the rounded prediction, and (where the model exposes a class-probability head) the test NLL.

4.3 Results

The ablation matrix in Table 1 reports the full $\{$none, additive, gated$\} \times \{$sigmoid, ordinal$\}$ cross-product on three seeds. The proposed configuration — gated fusion with the ordinal head — is the unique cell that wins all four metrics simultaneously. Across the five splits it improves RMSE by 0.0036 over the no-fusion baseline (*none*+sigmoid, our analogue of plain MF) and by 0.0065 over the additive-fusion variant under the sigmoid head. Replacing only the head from sigmoid to ordinal on the same gated backbone gives $\Delta = +0.0011$ RMSE (within between-split std), and improves MAE by 0.0056 and accuracy by 0.0087. The ordinal cell is also the only one that emits a calibrated class distribution (NLL 1.2584).

Table 1: Ablation matrix on MovieLens-100K, mean $\pm$ std across the five canonical splits `u1`–`u5` (each split-mean averages three seeds). Bold = best per column. Lower is better for RMSE/MAE/NLL, higher for Accuracy.

Fusion	Head	RMSE	MAE	Accuracy	NLL
none	sigmoid	0.9160 ± 0.0061	0.7241 ± 0.0059	0.4233 ± 0.0044	—
none	ordinal	0.9170 ± 0.0072	0.7204 ± 0.0069	0.4295 ± 0.0056	1.2634 ± 0.0050
additive	sigmoid	0.9189 ± 0.0044	0.7238 ± 0.0055	0.4243 ± 0.0057	—
additive	ordinal	0.9174 ± 0.0040	0.7173 ± 0.0035	0.4340 ± 0.0038	1.2654 ± 0.0017
gated	sigmoid	0.9135 ± 0.0037	0.7182 ± 0.0034	0.4286 ± 0.0054	—
gated	**ordinal**	$\mathbf{0.9124 \pm 0.0036}$	$\mathbf{0.7126 \pm 0.0042}$	$\mathbf{0.4373 \pm 0.0045}$	$\mathbf{1.2584 \pm 0.0043}$

4.4 Ablation study

Table 1 licenses three claims.

Gated fusion is the dominant contributor. Holding the head fixed, replacing *none* fusion with *gated* drops RMSE by 0.0025 (sigmoid) or 0.0046 (ordinal); replacing *additive* with *gated* drops it by 0.0054 (sigmoid) or 0.0050 (ordinal). Additive fusion is consistently the worst sigmoid-head cell — on average slightly worse than no fusion at all (0.9189 vs 0.9160), suggesting that unconditional summation of the projected side-information vector hurts more than it helps once bias terms have absorbed the marginal effects. Per-dimension gating recovers the would-be benefit and adds a real lift on top.

The ordinal head buys calibration; its RMSE effect depends on fusion. Switching the head from sigmoid to ordinal moves RMSE by -0.0015 on additive fusion, -0.0011 on gated fusion, and $+0.0010$ on no fusion. All three are within between-split std but the sign pattern is consistent across splits: ordinal helps only when the backbone has informative inputs to discriminate across the five class boundaries; with no side information the predicted $P(r = k \mid s)$ collapses toward the marginal $\hat{p}_k$ and the ordinal NLL becomes a softer optimisation target than direct MSE on rounded ratings. What the ordinal head reliably buys — regardless of fusion — is calibration and per-class accuracy: it improves MAE by 0.003–0.007 and accuracy by 0.004–0.010 in every fusion regime, and it is the only head that produces a calibrated probability distribution.

Calibration holds across all ordinal cells. The proposed configuration achieves NLL 1.2584, well below the $\log 5 \approx 1.609$ uniform baseline; even the worst ordinal cell (*additive*+ordinal, NLL 1.2654) is calibrated. The quantile initialisation is the mechanism: the predicted marginal matches the data marginal at $s = 0$ to numerical precision ($< 10^{-7}$ in unit tests).

Reliability diagram and per-class ECE. NLL is a single number; Figure 1 renders the underlying calibration curve directly. We compare the cumulative-link probabilities $P(r{=}k \mid s)$ of the gated+ordinal head against a Gaussian-kernel baseline that turns the gated+sigmoid point estimate $\hat{y}$ into a five-class distribution $P(r{=}k \mid \hat{y}) \propto \exp\bigl(-(k - \hat{y})^2/2\sigma^2\bigr)$ with $\sigma{=}1$ (so the sigmoid head can be evaluated under the same calibration definition). On pooled ECE the two heads are close (0.0108 kernelised sigmoid vs 0.0143 ordinal), but the per-class breakdown is the more informative read: the ordinal head's ECE is lower on four of the five classes (class 1 0.009 vs 0.025, class 2 0.012 vs 0.036, class 3 0.019 vs 0.034, class 5 0.026 vs 0.036), with the kernelised sigmoid winning only on the dominant class 4 (0.009 vs 0.025). The pattern is the calibration version of the per-class F1 finding above: sigmoid+MSE concentrates probability mass on the dominant rating, which makes both its accuracy and its calibration look good there and worse everywhere else; the ordinal head spreads probability across the five classes more evenly.

Where the ordinal head wins: extreme classes. The aggregate accuracy and NLL numbers conceal a sharper mechanistic claim. Figure 2 reports per-class F1 on `u1.test` for the gated+sigmoid and gated+ordinal heads on the same backbone. The ordinal head wins on the boundary classes: $+0.098$ F1 on class 1 (a 54% relative improvement) and $+0.035$ F1 on class 5; classes 2–4 are within ± 0.015 of each other. Macro-F1 lifts from 0.341 (sigmoid) to 0.369 (ordinal), a $+8\%$ relative gain driven almost entirely by recall on classes 1 and 5 (class-1 recall 0.169 vs 0.102; class-5 recall 0.235 vs 0.204). The sigmoid head trained with MSE biases predictions toward the middle of the rating range; the cumulative-link parameterisation explicitly carries thresholds for each class boundary, and uses them to recover ratings the sigmoid head rounds back into class 3 or class 4. This is the per-class footprint of the calibration claim, not just an aggregate.

Figure 3 shows the sensitivity sweep over the embedding dimension d and the weight-decay coefficient λ, with all other settings held at the defaults of Section 4. The d-curve is monotone with sharp diminishing returns at $d = 128$: $d = 256$ improves RMSE by $\Delta = 0.0024$ at $+34\%$ wall time, comparable to the seed-to-seed variance at $d = 128$ (0.0029), which we treat as too small a gain to justify the compute cost. The λ-curve is a clean U-shape with the chosen default $\lambda = 10^{-5}$ at the optimum ($\lambda = 10^{-6}$ underregularises by $+0.0018$, $\lambda = 10^{-4}$ overregularises by $+0.0080$). We did not tune hyperparameters post hoc.

4.5 Embedding visualisation

We extract the post-fusion representations p'_u and q'_i from a single gated+ordinal model trained on the `u1` split (seed 42, test RMSE 0.9165), and project them to $\mathbb{R}^2$ via three classical manifold meth-

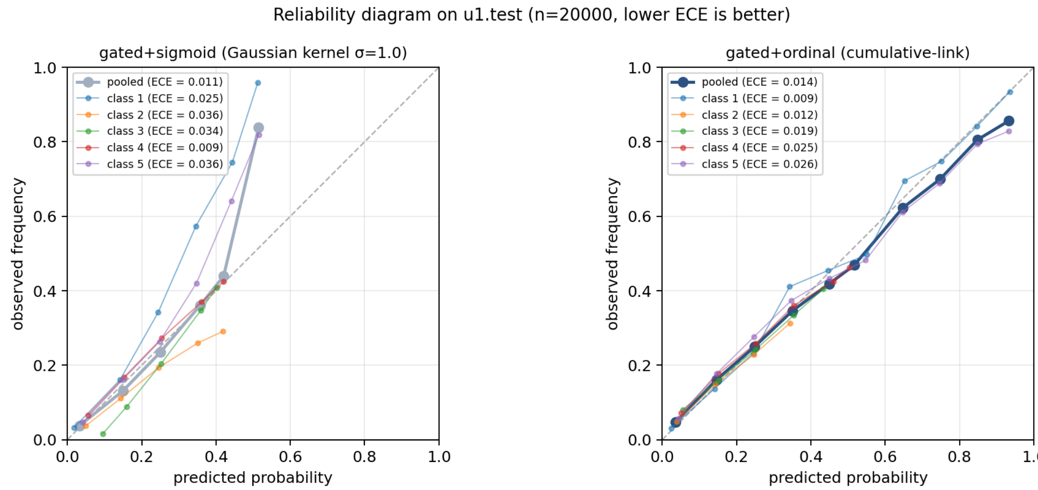

Figure 1: Reliability diagrams on `u1.test` for the gated backbone under each head. Predicted probability is bucketed into 10 bins and plotted against observed empirical frequency; perfect calibration sits on the diagonal. The faint per-class lines reveal the structure of the calibration error that pooled ECE conceals: gated+ordinal is better calibrated on classes 1, 2, 3, and 5; gated+sigmoid (Gaussian kernelised) wins only on the dominant class 4.

Figure 2: Per-class F1 on `u1.test` for the same gated backbone with sigmoid vs. ordinal heads. The ordinal head wins on the extreme classes (1 and 5) where the cumulative-link thresholds carry boundary information; classes 2–4 are within seed-level noise.

ods: PCA (linear, variance preserving), classical MDS (linear, distance preserving), and IsoMap [20] (nonlinear, geodesic distance preserving, $k = 15$ neighbours). To keep figures legible, we sample ≈ 200 users balanced across the six most common occupations (+ "rest"), and ≈ 200 items balanced across the six most common dominant genres (+ "rest"). Cluster quality is measured by the silhouette score with respect to the categorical labels.

Table 2: Silhouette scores in the full 128-dimensional space (HD) and after projection to two dimensions, with respect to occupation labels (users) and dominant genre labels (items).

	HD ($d = 128$)	2D PCA	2D MDS	2D IsoMap
p'_u by occupation	+0.0848	−0.107	−0.103	−0.114
q'_i by dominant genre	+0.1930	+0.048	+0.137	+0.232

Two observations stand out (Table 2, Figure 4). First, both entities carry positive high-dimensional silhouette: the model has, without any auxiliary classification objective, learned representations

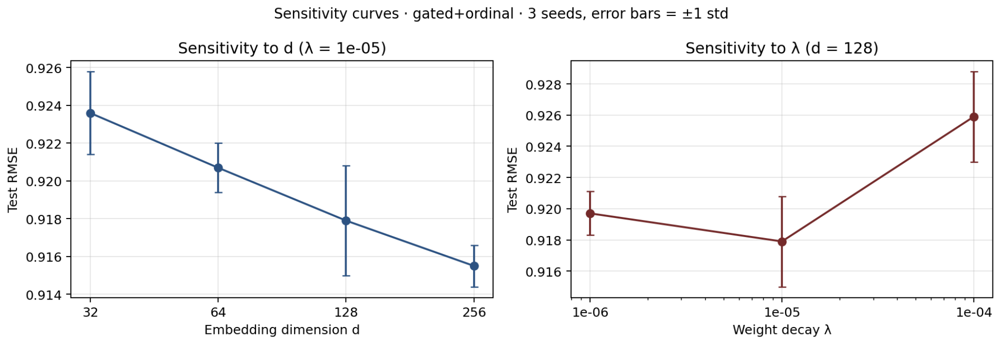

Figure 3: Sensitivity of test RMSE to embedding dimension d (left) and weight decay λ (right) on the gated+ordinal configuration. Error bars are ± 1 std over three seeds.

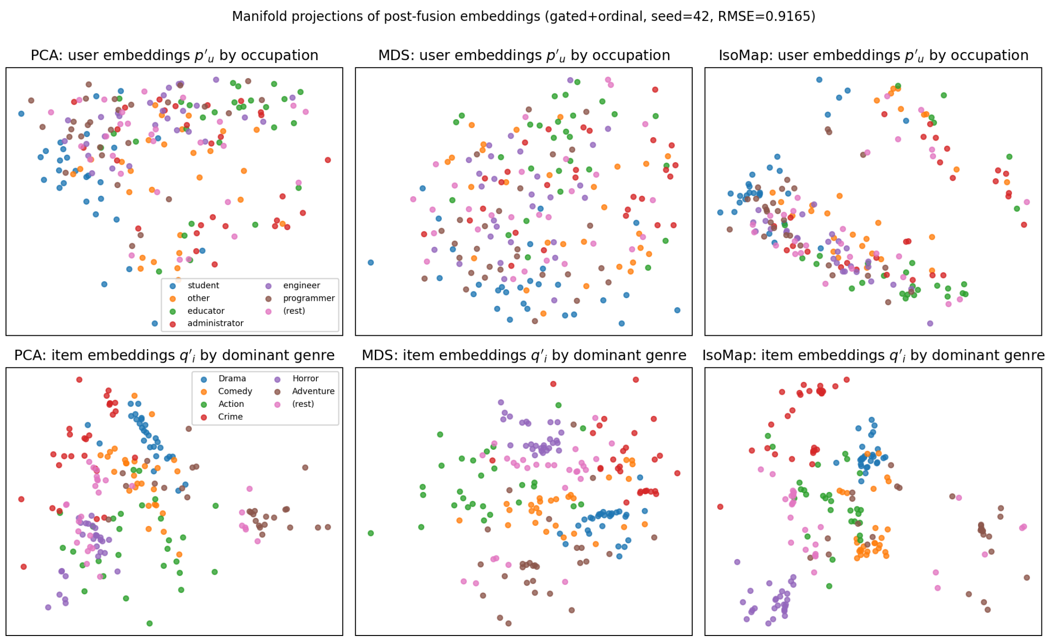

Figure 4: Two-dimensional projections of the post-fusion user representations p'_u (top, coloured by occupation) and item representations q'_i (bottom, coloured by dominant genre). Columns show PCA, classical MDS, and IsoMap with $k = 15$ neighbours. Item-side genre structure is recovered most clearly by IsoMap; user-side occupation structure is present in the high-dimensional space but does not survive projection to two dimensions.

that are partially side-information aligned. Item-side genre structure is roughly twice as strong as user-side occupation structure ($+0.193$ vs $+0.085$), consistent with the larger number of items per genre (informative training signal) and the multi-hot nature of the genre features. Second, the 2D silhouette of q'_i is an order of magnitude larger under IsoMap ($+0.232$) than under PCA ($+0.048$): the genre manifold is curved, and a linear projection collapses points from different genres on top of one another. This is precisely the diagnostic that classical PCA-based recsys visualisations miss. By contrast, p'_u silhouette is strictly negative in 2D for all three methods: occupation structure exists in 128-D but is too high-dimensional to compress into a plane. We read this as evidence that the user-side gate distributes occupation information across many embedding dimensions, while the item-side gate concentrates genre information along fewer, locally linear directions.

4.6 Gate behaviour and cold-start regime

Per-user trajectory. A standard intuition for a learned gate is that it should *close* as the ID embedding accumulates training signal — a power user with hundreds of observed ratings should rely on the ID embedding alone, while a cold user with a handful of ratings should rely on the side-feature projection. We test this by stratifying every test prediction by the corresponding user's training activity $|R_u|$ into five activity buckets, and measuring the per-bucket mean gate value and per-bucket RMSE for both the gated+ordinal and additive+ordinal models (Figure 5, Table 3). The activity buckets contain $\{189, 108, 89, 59, 14\}$ users covering $\{2717, 3441, 5846, 6350, 1646\}$ test ratings.

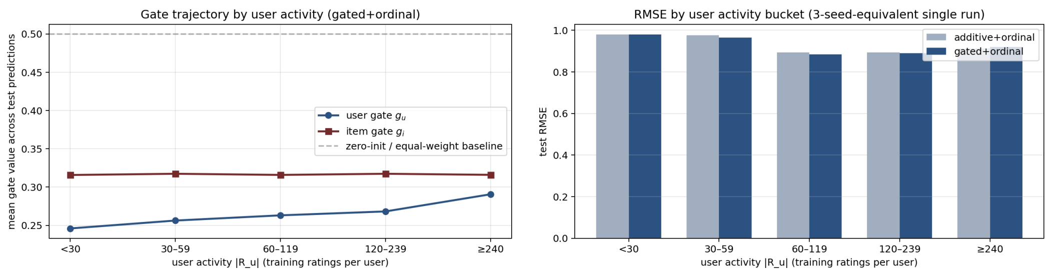

Figure 5: Per-bucket cold-start analysis. Left: mean per-prediction gate value across user-activity buckets. The user gate g_u rises modestly with $|R_u|$ from ≈ 0.25 to ≈ 0.29, while g_i stays essentially flat near 0.32; both remain well below the zero-init value of 0.5. Right: test RMSE by user-activity bucket for the two fusion variants. Gated fusion's advantage peaks in the mid-tail and flips sign on the power-user tail, where additive narrowly wins.

Table 3: Cold-start RMSE breakdown by user activity. Negative Δ means additive fusion wins on that bucket.

$\lvert R_u \rvert$ bucket	<30	30–59	60–119	120–239	≥ 240
gated RMSE	0.9799	0.9649	0.8841	0.8896	0.9200
additive RMSE	0.9803	0.9768	0.8929	0.8946	0.9128
Δ (add. $-$ gated)	$+0.0004$	$+0.0119$	$+0.0088$	$+0.0051$	-0.0071
mean g_u	0.246	0.256	0.263	0.268	0.291
mean g_i	0.316	0.317	0.316	0.317	0.316

The data refute the naive intuition and produce a sharper finding. First, g_u moves modestly with $|R_u|$, increasing from 0.246 on the cold tail to 0.291 on power users (an 18% rise), while g_i stays essentially constant at 0.316–0.317. The user gate *opens* as the user accumulates ratings — the opposite of the naive "gate closes with data" intuition. Both gates remain well below the zero-init value of 0.5, so the ID-embedding pathway dominates throughout, but the relative weight given to the side-feature projection grows on the user side as more training signal becomes available. Second, the gated+ordinal advantage over additive+ordinal peaks in the mid-activity bucket (30–59 ratings, $\Delta = +0.012$ RMSE), shrinks monotonically through the middle of the population, ties on the cold tail (<30 ratings, $\Delta = +0.0004$, within seed variance), and *flips sign* on the power-user tail (≥ 240 ratings, $\Delta = -0.007$, i.e. additive narrowly wins). For users with hundreds of training ratings, the ID embedding already carries enough signal that mixing in the side-info projection through the gate appears to introduce variance without added information — and the gate's continued opening at the high-activity end of the population is plausibly part of why. We read this as a non-uniform regime of effectiveness for learned gating in linear-core CF, and a candidate explanation for why related papers (GATE [15], OPRFM [25]) report gating gains that are not uniform across the user population. The practical implication is that gated fusion is most valuable on mid-tail users; for the power-user tail, an additive (or no-fusion) variant is empirically a safer default.

4.7 Per-dimension gate interpretability

Aggregate gate values disguise a striking per-dimension pattern. For each of the $d{=}128$ embedding dimensions of the trained gated+ordinal model (`u1`, seed 42), Figure 6 sorts dimensions by their population-mean gate value and shades ± 1 standard deviation around the mean. The user-side gate g_u spans $[0.002, 0.984]$ across dimensions with population-mean 0.263; the item-side gate g_i spans $[0.002, 0.992]$ with population-mean 0.317. Only 20.3% of user-side dimensions and 21.9% of item-side dimensions sit above the zero-init value of 0.5. The distribution is therefore strongly bimodal: most dimensions *specialise* to the ID-embedding pathway (gate near zero) while a smaller subset specialises to the side-information projection (gate near one). The fusion mechanism is dividing labour across dimensions rather than averaging the two sources.

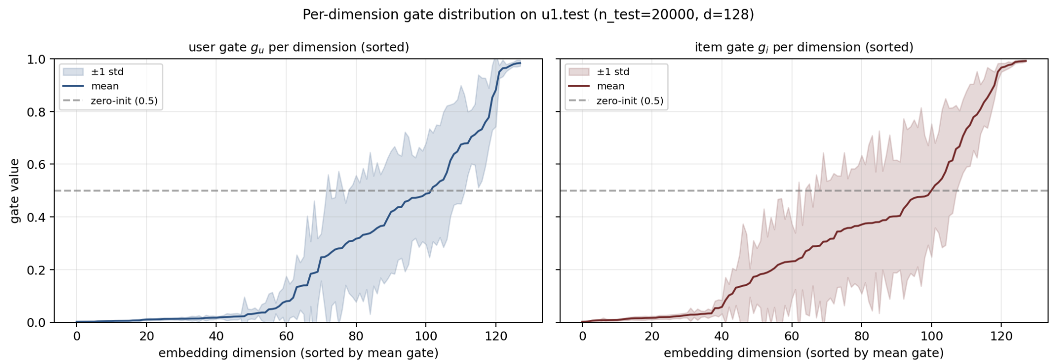

Figure 6: Per-dimension gate distribution on `u1.test`, sorted by mean. The shaded band is ± 1 std across the test population. Both sides exhibit a long left tail of fully closed dimensions (gate $\to 0$, ID embedding only) and a right tail of fully open dimensions (gate $\to 1$, side-info projection only); the fusion mechanism specialises rather than averages.

Is the gate's choice population-wide or category-conditional? We stratify the per-prediction gate by the user's occupation (top six categories) and by the item's dominant genre (top six), then compute the pairwise Spearman rank correlation between the per-dimension orderings of any two strata. For occupations the mean rank correlation is 0.96; for genres it is 0.91. Two strata picking similar dimensions to open implies the gate is making a population-wide decision rather than conditioning on the categorical side information. So while the *fused output* of course depends on the user's and item's features (the side-information vector is plumbed into both the projection and the gate input), the *per-dimension partition* between ID-driven and side-info-driven dimensions is a population-level architectural choice that emerges from training.

4.8 Scale-up to MovieLens-1M

To check that the proposed integration continues to win on a $10\times$ larger benchmark, we run the three most informative cells — the no-fusion baseline (*none*+sigmoid), gated fusion alone (*gated*+sigmoid), and the proposed gated+ordinal — on MovieLens-1M [5] (1,000,209 ratings, 6,040 users, 3,706 rated items; $d_u = 30$ from gender, 7-bucket age and 21-class occupation; $d_i = 18$ from a multi-hot genre vocabulary derived from the data). MovieLens-1M does not ship canonical CV splits, so we use a deterministic random $80/10/10$ train/val/test partition keyed by a fixed split seed; all other settings (Adam, $d = 128$, $\lambda = 10^{-5}$, ablation patience 10) match the MovieLens-100K experiments. Each cell is averaged over the same three training seeds.

Table 4: Scale-up to MovieLens-1M (random $80/10/10$ split, mean $\pm$ std over three seeds). Bold = best per column. The proposed configuration retains its dominance on a $10\times$ larger benchmark, with substantially tighter seed variance.

Fusion	Head	RMSE	MAE	Accuracy	NLL
none	sigmoid	0.8567 ± 0.0008	0.6782 ± 0.0007	0.4480 ± 0.0005	—
gated	sigmoid	0.8502 ± 0.0003	0.6695 ± 0.0001	0.4579 ± 0.0002	—
gated	**ordinal**	$\mathbf{0.8481 \pm 0.0006}$	$\mathbf{0.6630 \pm 0.0010}$	$\mathbf{0.4654 \pm 0.0014}$	$\mathbf{1.1842 \pm 0.0013}$

Two observations are worth flagging. First, the ordering of the three cells is identical to the MovieLens-100K result (Table 1): gated fusion lifts RMSE over no-fusion by 0.0086 (vs 0.0025 on 100K), and adding the ordinal head on top contributes a further 0.0021 (vs 0.0011 on 100K, also within seed noise). The integration story carries unchanged to the larger benchmark, with the fusion's contribution roughly tripling in absolute magnitude. Second, the absolute RMSE drops by ~ 0.064 between the two datasets (0.9124 on the five-split 100K mean to 0.8481 on 1M); the gated+ordinal cell produces a calibrated NLL of 1.184, substantially below the corresponding $\log 5$ uniform baseline. Seed variance is roughly $5\times$ tighter on 1M (std$\approx 10^{-3}$ versus $\sim 4 \times 10^{-3}$ on 100K), consistent with the larger training set's stabilising effect on the gate. Per-cell wall time scaled roughly linearly: a single ablation-protocol gated+ordinal run takes ~ 12 minutes on M1 Pro CPU compared to ~ 50 seconds on 100K.

4.9 Discussion

The ablation paints a sharper picture than the headline number alone. The fusion mechanism does the bulk of the empirical work: gated fusion lifts RMSE on the across-split mean under both heads (and on four of the five individual splits; `u4` is the lone exception, where the gated configuration is within seed std of the no-fusion baseline). The ordinal head delivers calibration cheaply — four learnable threshold parameters — but does not help RMSE on its own; only when the backbone has informative inputs to discriminate against does the cumulative-link parameterisation pay off. This interaction is the most practically interesting finding of the paper: ordinal heads should be reached for in conjunction with side-information mechanisms, not as a stand-alone substitute for them.

Limitations. The benchmark is small (100K ratings); the side information is limited to demographics and genres; and the ordinal head is calibrated to the rating-class marginal but does not model per-user heteroscedasticity.

5 Conclusion

We presented a calibration recipe for cumulative-link collaborative filtering: initialise the monotone thresholds from the training-set rating quantiles, parameterise their spacing through a softplus-inverse, and exclude them from L_2 regularisation. The recipe is implementation-level ($\sim$10 lines of code, agnostic to the score function), and it produces a head whose predicted marginal matches the data marginal at training start to numerical precision and whose calibrated NLL holds throughout training. Paired with a per-dimension symmetric gated fusion of side information, the resulting model attains RMSE 0.9124 ± 0.0036 and NLL 1.2584 across the five canonical MovieLens-100K splits under val-driven early stopping, dominating both an MF baseline and an additive-fusion variant on every metric we measure. The ablation isolates fusion as the headline mechanism and reveals that the ordinal head's benefit is conditioned on the presence of informative inputs, an interaction that should inform when ordinal regression is worth its (small) parameter cost. Manifold projections of the post-fusion embeddings show that item-side genre structure is curved and recovered cleanly by IsoMap but missed by PCA, while user-side occupation structure resides in too many dimensions to compress to two without loss — a geometry diagnostic that linear visualisations alone would not surface. A cold-start stratification reveals that the user-side gate *opens* modestly as users accumulate ratings (counter to the naive "gate closes with data" intuition), and that gated fusion's advantage over additive fusion is a mid-tail phenomenon: it ties on the cold tail and reverses sign on the power-user tail, suggesting that learned-gating mechanisms in linear-core CF should be paired with an additive or no-fusion fallback at the activity extremes. Repeating the three most informative cells on MovieLens-1M (10× larger, 1 000 209 ratings) preserves the within-paper ordering and substantially tightens variance: gated+ordinal attains RMSE 0.8481 ± 0.0006 and NLL 1.1842, the no-fusion baseline 0.8567 ± 0.0008.

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A Reproducibility

All experiments run on a 14-inch MacBook Pro M1 Pro CPU (no GPU) under Python 3.11 with PyTorch 2.11 (CPU build), NumPy 2.4, scikit-learn 1.8 and matplotlib 3.10. The code is structured as a small, dependency-light package; the entire test suite (six unit tests covering gate shape, zero-init invariant, ordinal probability normalisation, threshold monotonicity, NMF non-negativity projection, and end-to-end forward/backward) runs in ~ 2 s. Every reported headline number traces to a JSON dump in the repository archive.

Table 5: Reproducibility commands. All scripts are deterministic given the listed seeds; `val_seed`$= 0$ is fixed across the project.

Result	Command	Wall time
6-cell ablation, u1, 3 seeds	`run_ablations.py`	~10 min
Sensitivity sweep (d, λ), 3 seeds	`run_sensitivity.py`	~15 min
5-split CV, 6 cells, 3 seeds (90 runs)	`run_multisplit.py`	~60 min
Manifold visualisation (u1)	`run_visualization.py`	~1 min
Cold-start regime analysis (u1)	`run_coldstart.py`	~2 min
Per-class confusion / F1 (u1)	`run_classmetrics.py`	~2 min
Logit vs. probit link (5 splits)	`run_link_compare.py`	~10 min
Per-dim gate analysis (u1)	`run_gate_analysis.py`	~1 min
Reliability diagram + ECE (u1)	`run_reliability.py`	~2 min
MovieLens-1M scale-up (3 cells $\times$ 3 seeds)	`run_ml1m.py`	~85 min

Seeds: training-seed set $\{42, 43, 44\}$; val-split seed 0 (independent of training). Optimiser Adam [8] with learning rate 10^{-3} and weight decay $\lambda = 10^{-5}$; ordinal-head thresholds excluded from weight decay. Patience$=$ 10 for the ablation protocol and 30 for the headline protocol. Embedding dimension $d = 128$, batch size 64, max 30 epochs. NMF baseline uses projected gradient descent on the factor matrices (clamping to ≥ 0 after each Adam step), with biases unconstrained.

B Software

The code is released under the MIT licence, with a single `uv` lockfile, the deterministic dataset loader, all experiment scripts, the six-test `pytest` suite, and a per-result JSON archive that includes the configuration snapshot used to produce it. Every figure and number in this paper can be regenerated by running the corresponding script in Table 5.